

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular and Supplementary Summer 2024 Course: B. Tech. Branch: Electronics and Computer Engineering / Electronics and Computer Science Engineering Semester : IV Subject Code & Name: BTBS404 Probability Theory and Random Processes Max Marks: 60 Date: 20/06/2024 Duration: 3.00 Hr.																	
Instructions to the Students: 1. All the questions are compulsory. 2. Use of non-programmable scientific calculators is allowed. 3. Assume suitable data wherever necessary and mention it clearly.																	
		(Level/CO)	Marks														
Q. 1	Solve Any Two of the following.		12														
A)	A bag contains 7 white, 6 green and 5 black balls. Two balls are drawn at random. Find the probability that they will both be white.	CO1	6														
B)	A Problem on calculus is given to two students Seeta & Geeta whose chances of solving it are $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability of the problem being solved, if both of them try independently.	CO1	6														
C)	If a machine is set up correctly it produce 90% good items, if it is incorrectly set up then it produce 10% of good items. Chances for a setting to be correct & incorrect are in the ratio 7:3 after a setting is made the first two items produced are found to be good items. What is the chance that the setting was correct?	CO1	6														
Q.2	Solve Any Two of the following.		12														
A)	Consider that the lifetime(x) & the brightness (Y) of a light tube are continuous random variables. The joint probability density function $f_{xy}(X,Y) = \lambda_1\lambda_2e^{-(\lambda_1X+\lambda_2Y)}$, $0 < X < \infty, 0 < Y < \infty$, than find marginal density functions.	CO2	6														
B)	The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped, find the probability that i) Exactly two will strike the target. ii) At least two will strike the target.	CO2	6														
C)	The life of army shoes is normally distributed with mean eight months & S.D. two months. If 5000 pairs are issued, how many pairs would be expected to need replacement after 12 months? (The area under standard normal curve between 0 to 2 is 0.4772)	CO2	6														
Q. 3	Solve Any Two of the following.		12														
A)	Calculate the coefficient of correlation by Karl Pearson's method for following table	CO3	6														
<table><tr><td>X</td><td>10</td><td>14</td><td>18</td><td>22</td><td>26</td><td>30</td></tr><tr><td>Y</td><td>18</td><td>12</td><td>24</td><td>6</td><td>30</td><td>36</td></tr></table>				X	10	14	18	22	26	30	Y	18	12	24	6	30	36
X	10	14	18	22	26	30											
Y	18	12	24	6	30	36											

B)	If $r = 0.8$, $\sum xy = 60$, $\sigma_y = 2.5$ & $\sum x^2 = 90$ then find the number of items (n). Where x and y are the deviation from the arithmetic mean.	C03	6																						
C)	Obtain the Spearman's rank correlation coefficient to the following data <table border="1"> <tr> <td>X</td> <td>70</td> <td>64</td> <td>82</td> <td>55</td> <td>64</td> <td>90</td> <td>82</td> <td>50</td> <td>60</td> <td>64</td> </tr> <tr> <td>Y</td> <td>70</td> <td>62</td> <td>75</td> <td>52</td> <td>85</td> <td>65</td> <td>75</td> <td>45</td> <td>60</td> <td>80</td> </tr> </table>	X	70	64	82	55	64	90	82	50	60	64	Y	70	62	75	52	85	65	75	45	60	80	C03	6
X	70	64	82	55	64	90	82	50	60	64															
Y	70	62	75	52	85	65	75	45	60	80															
Q.4	Solve Any Two of the following.		12																						
A)	If θ is the acute angle between the two regression lines for two variables X and Y, then show that $\tan\theta = \left \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{r^2 - 1}{r} \right) \right $	C04	6																						
B)	Two lines of regressions are $5y-8x+17=0$ and $2y-5x+14=0$. If $\sigma_y^2 = 16$ find i) The mean values of x and y. ii) The variance of x. iii) Correlation coefficient between x and y.	C04	6																						
C)	If $\bar{x}=8.2$, $\bar{y}=12.4$, $\sigma_x = 6.2$, $\sigma_y = 20$, $r(x, y) = 0.9$ then find the lines of regression. Estimate the value of x for y = 10 & estimate the value of y for X = 10.	C04	6																						
Q. 5	Solve Any Two of the following.		12																						
A)	In a random sample of 340 students, 178 of the 210 females and 90 of the 130 males passed Statistics and Probability on their first take. Construct a 90% confidence interval for the population proportion of students who passed the subject.	C05	6																						
B)	A stenographer claims that she can take dictation at the rate of 120 words per minute. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words with a standard deviation of 15 words ? Use 5 percent level of significance.	C05	6																						
C)	700 ladies, out of sample of 1000, were consumers of lipsticks before GST. After GST 600 ladies, were consumers of lipsticks in a sample of 900 persons. Find out whether there is significant decrease in the consumption of lipsticks after GST. Given 1 % level of significance $Z_\alpha = 2.33$	C05	6																						
*** End ***																									